Program: BE

Date	10.11.2025	Maximum Marks	65
Course Code	CS235AI	Duration	120 Min
Sem	III	CIE – I	

OPERATING SYSTEMS

(Common to CSE/CD/CY/ISE)

Sl. No.	Quiz Questions	M	L	CO
1	Identify the multithreading model where i. The entire process blocks when a thread makes blocking system call ii. The entire process does NOT blocks when a thread makes blocking system call	**02**	L3	CO3

2	Analyze the following code segment and predict the output.
	#include <stdio.h>
	#include <unistd.h>
	#include <sys/wait.h>
	int main() {
	  printf("A");
	  pid_t p1 = fork();
	  if (p1 == 0) {
	    printf("B");
	    fork();
	    printf("C");
	    exit(0);
	  } else {
	    wait(NULL);
	    printf("D");
	  }
	  printf("E");
	  return 0;
	}
		02	L4	CO2

3	For each of the following scenarios, identify the current process state and the state transition that will occur next: i. A process is executing on the CPU when a higher-priority process becomes ready. ii. A process is waiting for an I/O operation and the I/O request completes.	**02**	L3	CO3
4	Categorize the following operations as Privileged or Unprivileged: i. Changing the address space of a running process ii. Spawning a child process using fork() iii. Reading the thread ID of the current thread iv. Executing a context switch between two different processes	**02**	L3	CO3

5	Identify the memory segment where each variable and string literal is stored during program execution.
	
	char msg[] = "OS Quiz";
	char *ptr = "Memory";
	int main() {
	   register int r = 10;
	   int *arr = malloc(sizeof(int) * 3);
	}
		02	L3	CO2

Sl. No.	Test Questions	M	L	CO
1a	Write a multithreaded program in C to perform concurrent operations on an array using two threads. Thread-1 Counting even and odd numbers while Thread-2 Calculating the average of array elements. Execute both tasks simultaneously and display the results.	**10**	L3	CO4
2a	A user application needs to access a printer and allocate memory while running. If applications could directly interact with hardware or use privileged instructions, they might corrupt memory, crash the system, or compromise security. From this scenario, identify and explain the operating system mechanism that prevents such direct access to critical resources. How does it work, and what could happen if this protection were absent? Include a neat diagram showing the transitions involved.	**06**	L3	CO2
2b	With neat diagram discuss the mechanism of context switch between processes in operating system.	**04**	L2	CO1
3a	A data analytics system uses process creation to speed up computation. Write a C program where the parent process collects system data and then creates two child processes using fork(). Child Process 1 should analyze the dataset to determine and display the highest CPU usage recorded, while Child Process 2 should calculate and display the average CPU usage. Each child must print its PID, perform its computation, display the results, and then exit. The parent process should wait for both children to finish, print a confirmation message indicating their termination, and finally display “Parent Process Exiting…”.	**10**	L3	CO4
4a	Brief Amdahl's Law in the context of multi threaded programming. Given that 60% of a program's code can be parallelized and 40% must be executed sequentially, calculate the theoretical speedup when utilizing 4 threads/cores for parallel execution. Is Amdahl’s Law practical for real-world computing? Justify your answer.	**06**	L3	CO3
4b	Justify or contradict the following statements: i. The primary objective of multiprogramming is to minimize user response time, while the primary objective of time sharing is to maximize processor utilization. ii. A microkernel-based operating system is more efficient, reliable, secure, and flexible than a monolithic	**04**	L2	CO2
5a	Explain the various kernel architectures used in operating systems and explain how their components are organized and interact with each other.	**10**	L2	CO1

CO 1	Demonstrate the fundamental concepts of operating system like process management, file management, memory management and issues of synchronization.
CO 2	Analyze and interpret operating system concepts to acquire a detailed understanding of the course.
CO 3	Apply the operating systems concepts to address related new problems in computer science Domain.

CO 4	Design or develop solutions to solve applicable problems in operating systems domain.
CO5	Extend the theoretical knowledge acquired through the course to demonstrate skills like investigation, effective communication, working in team/Individual, following ethical practices by implementing operating system concepts/applications and engage in lifelong learning.

Course Outcomes

Blooms' taxonomy

L1	L2	L3	L4	L5	L6	CO1	CO2	CO3	CO4	CO5	CO6
	18	32				14	10	6	20		